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Courtship behaviour in the frog species, *Xenopus laevis*, involves male frogs calling to:

- attract sexually active females – these are advertisement calls
- start and continue mating – these are mating calls
- signal when a male is not sexually active – these are rasping calls.

Scientists investigated frog courtship behaviour by feeding a population of sexually active male frogs a diet containing the hormone EE2. The scientists also fed a separate control population of sexually active male frogs a diet without EE2.

They determined the percentage of males making advertisement calls or rasping calls in each population.

Table 6 shows their results.

Table 6

Population	Percentage of males making:	
	Advertisement calls	Rasping calls
Males fed EE2	94.0	4.0
Males not fed EE2 (control)	97.0	0.5

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There were 800 males in the control population.

Each male made one type of call.

Use this information and **Table 6** to calculate the number of males making mating calls in the control population.

[1 mark]

Answer _____ males

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Suggest **one** change the scientists could make to both frog populations to increase the number of mating calls.

[1 mark]

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The scientists also investigated the effect on **female** frog courtship behaviour of feeding EE2 to male frogs.

Table 7 shows their results.

Table 7

Population	Median time females spent in courtship / s
Males fed EE2	8
Males not fed EE2 (control)	16

EE2 is contained in human contraceptive pills. Some EE2 is released in human urine and collects in sewage. Untreated sewage pollutes the water in frog habitats.

Suggest and explain the effect EE2 pollution in frog habitats will have on frog breeding.

Use information from **Table 6** and **Table 7** in your answer.

[4 marks]

Effect on frog breeding _____

Explanation _____
